

Akshar Patel

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Active DoD Secret Clearance

EDUCATION

Rutgers University, New Brunswick, NJ

Expected Graduation: May 2028

B.S. in Aerospace Engineering (ABET)

- **GPA:** 3.68
- **Relevant Coursework:** Aerospace Materials, Dynamics, Dynamic Systems & Controls, Fluid Mechanics, Statics, Thermodynamics

TECHNICAL SKILLS

CAD: PTC Creo, Fusion 360, Siemens NX, SolidWorks, Onshape, AutoCAD

Analysis / Simulation: Fusion Simulations, Ansys Mechanical/CFD/ Fluent, Finite Element Analysis, Thermal Desktop

Programming: MATLAB/Simulink, Python, Java

Systems / Tools: Cameo, Atlassian, GitHub, Microsoft Office, Slack

EXPERIENCE

Lockheed Martin | Moorestown, NJ

Mechanical Engineering Apprentice

July 2026 - Present

- Ensured compliance with DSR6 regulations across 20+ drawings, validating multi-body parts in **PTC Creo** for manufacturing release
- Drove final fan placement by simulating forced convection from 3 fans in **Creo Simulate**, cutting peak board temperature by 10 C
- Led the initial thermal design by building a 1D **Excel** thermal-circuit model that mapped board heat gradients and traced conduction paths

Lockheed Martin | Moorestown, NJ

Program Management Apprentice

June 2026 - Present

- Co-authored the budget, schedule, and risk assessment package presented to a **business area president** in multiple formal gate reviews
- Cut **Jira** ticket entry time 96% by automating issue creation during Program Increment Planning, saving the team hours of manual task entry

Lockheed Martin | Moorestown, NJ

Systems Engineering Apprentice

June 2025 - June 2026

- Wrote 30 verifiable system requirements in **Cameo MBSE**, reducing integration risk across software and hardware interface boundaries
- Modeled system behavior across 10 **Cameo SysML diagrams**, translating stakeholder needs into a verified solution blueprint

Drifty | Remote

Co-Founder & CTO

January 2026 - Present

- Co-founded a sleep apnea intervention startup and led the building of software integrable on existing consumer wearable health devices
- Accepted into the LvlUp Ventures Labs Accelerator, **awarding us \$300k in AWS credits**; maintaining a \$100 per month burn rate
- Owned GTM strategy, leading campaigns that hit **1,000+ followers and 5,000+ views** at zero ad spend while shipping product iterations

Space Ocean Corp. | Remote

Lead Thermal Engineer

January 2026 – June 2026

- Developed a modular 6-node thermal model for a 12U CubeSat in **Python**, simulating transient temperature response over a full LEO cycle
- Validated a heater system to prevent water-tank freezing by running hot and cold case analyses against component survival temperature
- Conducted 5 **trade studies** on the thermal subsystem, scoring heater options against mass, power, and recurring cost to select a final design

Rutgers Formula Racing (FSAE) | Rutgers University

Aerodynamics Engineer

September 2024 – May 2025

- Designed aerodynamic components in **SolidWorks**, ensuring full compliance with FSAE safety and design regulations passing inspection
- Tested 5 aero components in **Ansys CFD**, using surface pressure distributions to trade drag against downforce and rank configurations
- Constructed 8 carbon fiber parts using wet-layup and resin infusion, running the layup schedules and vacuum bagging through full cure

PROJECTS

UAS Avionics Enclosure Structural & Thermal Qualification | *Fusion 360, FEA, Thermal Analysis, Iteration*

- Cleared 87 to 2,293 Hz **vibration band**, achieving +0.16 **margin of safety** by running multiple **modal FEA sweeps** in **Fusion 360**
- Proved sealed 10.8 W avionics enclosure needed 9.2x more cooling area, by conducting **Python heat transfer analysis**
- Sized an aluminum enclosure in **Fusion 360**, tracing all 15 structural and thermal **requirements** to component datasheets for verification

SpaceX Dragon Capsule Aero-thermal Re-entry Analysis | *Ansys Mechanical/CFD, Python*

- Built and validated capsule re-entry in **Ansys Fluent**, lining Rayleigh Pitot & modified Newtonian theory within 3% across Mach 5 to 10
- Stabilized hypersonic solver runs by iterating on the Courant number, then ramped Mach cases from a converged lower speed solution
- Compared 3 materials by coupling Fay-Riddell heating to a **Python** 1D transient conduction solver, finding Avcoat 43% heavier

FPV Drone Challenge | *Onshape, Iterative Design*

- Tuned a scratch-built FPV quadcopter to stable flight across 8 test flights, **integrating** motors, ESCs, and controller onto a carbon frame
- Expanded payload capability by designing and **3D Printing** 3 custom attachments validated in flight testing and iterating based on results
- Improved airframe durability by **diagnosing** crash **failure** modes and **revising** mount geometry over 3 rebuilds between testing sessions